MECH5200 Numerical Methods for PDEs : *Video 20: Finite Element Methods*

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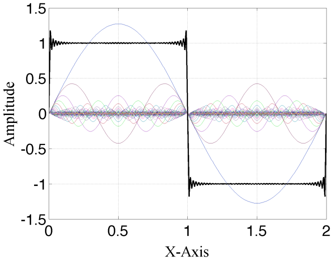
Let's Consider Fourier Analysis

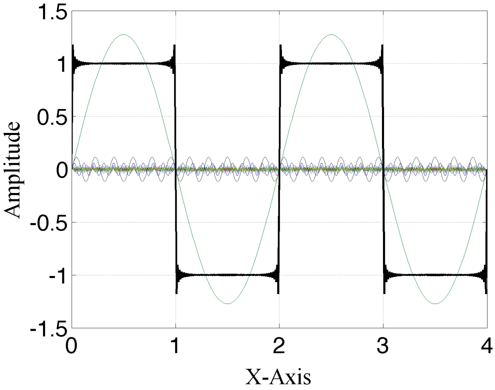
- The idea of Fourier analysis $\rightarrow$ construct the function using a series of orthogonal sine and/or cosine functions
- If different amounts of each harmonic are weighted and added together, you can represent different periodic functions:

$$f(x) = \sum_{n=1,3,5\ldots}^{\infty} \frac{4}{\pi n} \sin\left(\frac{n\pi x}{L}\right) \tag{1}$$



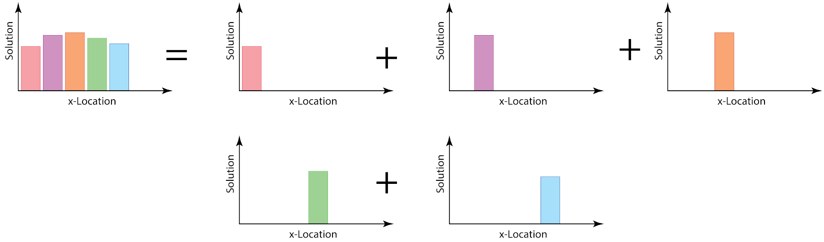
What's Wrong with Fourier Analysis?

- The sine and cosine functions have global support.
- ie. They exist across the domain
- Fixing boundary values $\rightarrow$ hard.



How do we fix this?

- **Recall** Our finite volume approach – for representation of the solution in the cell:
- Cell average value = solution in cell.
- Overall solution: Add together solutions from different cells.

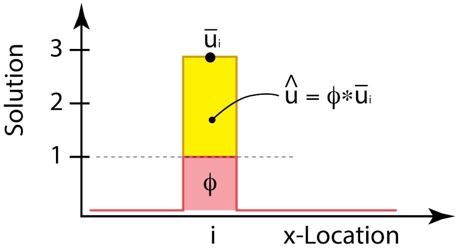


How do we fix this?

- Let's make this a little more abstract:
  - Define a constant basis function ϕ_i that satisfies the following:

$$\phi_i = \begin{cases} 1 & if (x_L < x < x_R) \\ 0 & otherwise \end{cases} \tag{2}$$

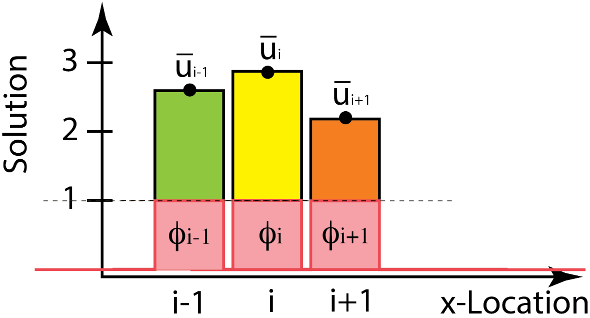
  - To construct the solution in a given cell, we can multiply ϕ_i by the cell average value: $\bar{u}_i$.



How do we fix this?

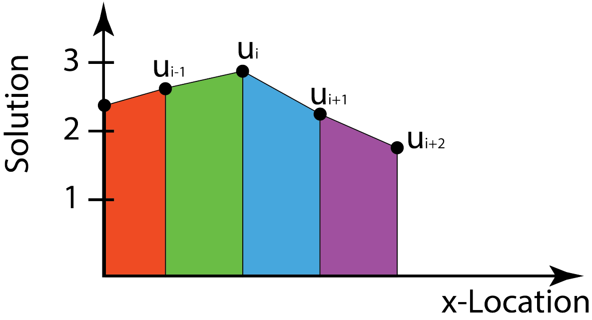
- To construct the approximate solution, $\hat{u}(x)$ across the whole domin, we sum the individual cell solutions:

$$\hat{u}(x) = \sum_{i=1}^{i=NV} \bar{u}_i \cdot \phi_i \tag{3}$$



Linear Basis Functions?

- **But....**
  - The solution has discontinuities/jumps (usually bad, but stay tuned)
  - Also, these constant basis functions don't represent solutions well
- Can we improve the solution representation?
  - Perhaps linear approximations across elements? How?

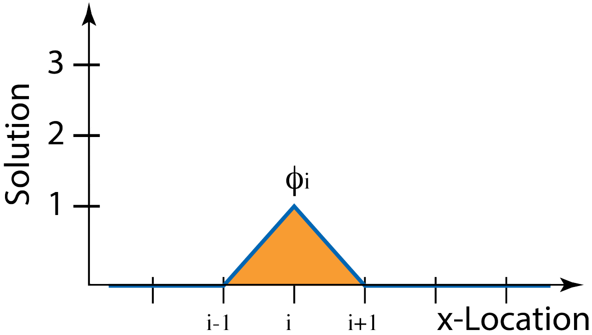


Linear Basis Functions

- Let's setup a **linear hat basis** function so that:

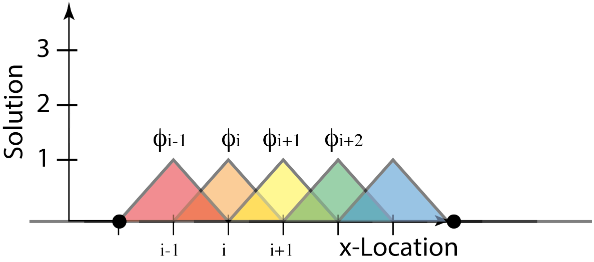
$$\phi_i = \begin{cases} 1 & at \;\; x = x_i \\ 0 & at \;\; other \;\; nodes \end{cases} \tag{4}$$

- The basis function varies linearly from node i to nodes $i-1$ and $i+1$:



Linear Basis Functions

1. Introduce a "linear hat" basis function at each node.
2. Scale the basis function by the value of the function at that node
3. Add the scaled bases together.



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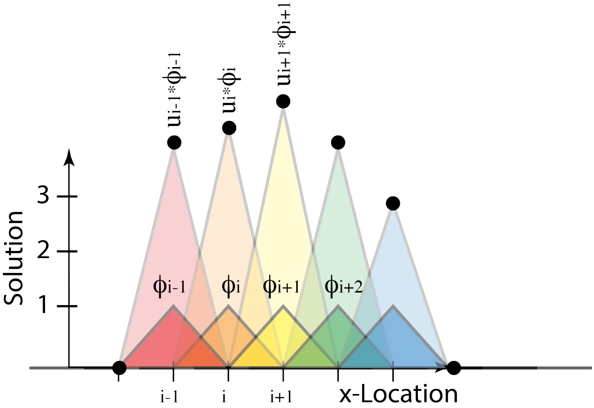

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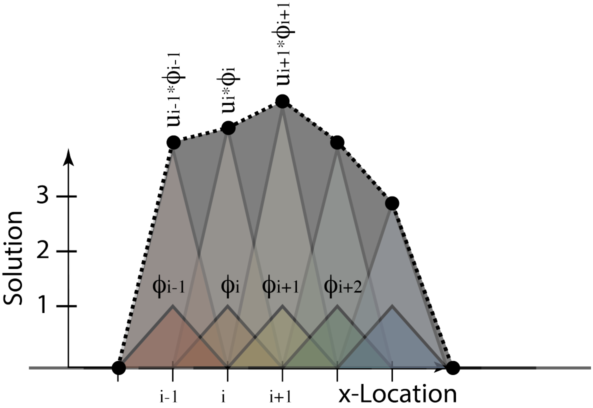
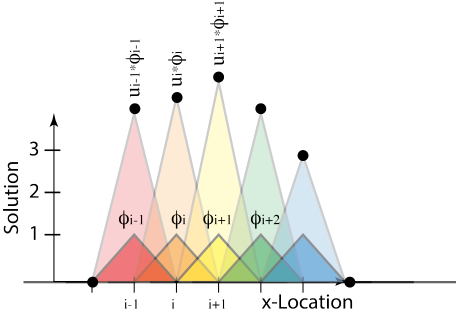


Linear Basis Functions

- This produces a general expression similar to the constant case examined earlier:

$$\hat{u}(x) = \sum_{i=1}^{NNodes} \phi_i(x) \cdot u_i \tag{5}$$

- Where $\phi_i(x)$ is the hat basis function.



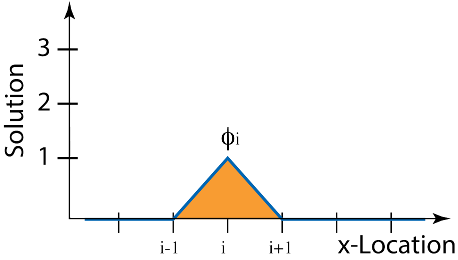
1-D Linear Hat Basis function

- The linear hat basis function spans two discrete elements in the domain.
- The basis function value associated with an arbitrary node i is:

$$\phi_i^+(x) = \frac{1}{h_n}(x_{i+1} - \xi) \qquad x_i < \xi < x_{i+1} \tag{6}$$

$$\phi_i^-(x) = \frac{1}{h_{n-1}}(\xi - x_{i-1}) \qquad x_{i-1} < \xi < x_i \tag{7}$$

$$\phi_i = 0 \qquad \textit{elsewhere} \tag{8}$$



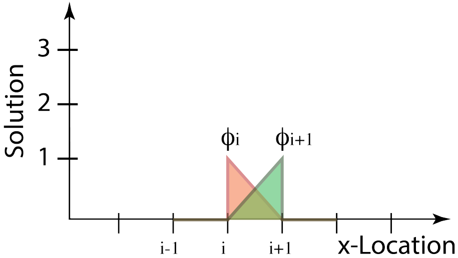
1-D Linear Hat Basis Function

- At times, it is simpler to analyze each element i independently:

$$\phi_{i+1}(\xi) = \frac{1}{h}(\xi) \qquad 0 < \xi < h \tag{9}$$

$$\phi_i(\xi) = \frac{1}{h}(h - \xi) \qquad 0 < \xi < h \tag{10}$$

$$\tag{11}$$



- ... and then save the information in terms of nodes.

What have we learned?

- Basis functions can be used to represent a solution.
- Summing the weighted basis functions can yield functional representation in domain:

$$\hat{u}(x) = \sum_{i=1,2,3...}^{NVals} u_i \cdot \phi_i(x) \tag{12}$$

- Using linear (or higher order) basis functions allows more accurate representation of basis.
- We developed expressions for linearly varying basis functions.